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Seasonal Decomposition-Enhanced Deep Learning Architecture for Probabilistic Forecasting

Keyan Jin[1] | Francisco Javier Blanco-Encomienda[2]

[1]International School for Postgraduate Studies, University of Granada, Granada, Spain | [2]Faculty of Education, Economy and Technology, University of Granada, Ceuta, Spain

Correspondence: Francisco Javier Blanco-Encomienda (jble@ugr.es)

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ABSTRACT

Time series decomposition as a general preprocessing method has been widely used in the field of time series forecasting. However, since the future is unknown, this preprocessing practice is limited in realistic forecasting, especially in real-time forecasting scenarios. In this paper, we propose a framework with time series decomposition and probabilistic forecasting capabilities. Distinguishing from models based on time series pre-decomposition, our proposed framework can decompose the series into trend components and seasonal components in real time to achieve end-to-end forecasting. We apply this framework to four state-of-the-art deep time series models and test their performance on four synthetic datasets and the WTI oil price dataset. The results show that the seasonal decomposition-based framework can significantly improve the point and probabilistic forecasting accuracy of the original models.

1 | Introduction

Forecasting financial time series is a challenging task for two main reasons: (1) the complex nonlinearity. When a source has both commodity and financial attributes, its price is not only affected by supply and demand but also greatly influenced by other factors such as gross domestic product, stock market activities, foreign exchange rates, and even weather conditions (Bernabe et al. 2004; Tang et al. 2012; Yousefi and Wirjanto 2004). These factors may lead to a high degree of nonlinearity. Moreover, in the absence of a clear nonlinear relationship, large computational costs are consumed for inference. (2) The complex temporal patterns. In general, time series can be regarded as a superposition of different mode functions or a combination of trend, seasonality and noise (Wen et al. 2021). The entanglement of multiple components may make the temporal pattern exhibit irregularity, reflected in unreliable and invisible dependencies. In response to these problems, researchers have developed many models over the past decades to capture the complex features of time series in order to effectively infer the future. In general, prediction models can be broadly classified into two categories: statistical models and machine learning models (Dastile et al. 2020; Jin et al. 2023). Statistical models rely on stronger assumptions between the data and the model to estimate the explicit model structure and thus forecast, such as ARIMA, SARIMA, and so forth. Although these models have achieved superior predictive performance in some applications, they are expensive to use because the model assumptions require strong domain knowledge. Moreover, statistical models have limited nonlinear representation capability and perform poorly especially with massive data and features. Machine learning can be understood as a large-scale optimization task that resembles a functional optimization problem in its formulation,

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but is complex due to its large scale. Common machine learning models include support vector machine (SVM), gradient boosting tree (GBDT), artificial neural network (ANN), and so forth. Among them, neural network-based deep learning (DL) techniques are widely studied and applied and gradually become popular methods in the field of time series forecasting, such as LSTM (Hochreiter and Schmidhuber 1997), DeepTCN (Chen et al. 2020), TPA (Shih et al. 2019), and LogTrans (Li et al. 2019a). Compared to classical statistical models, DL shows great potential in expressing complex nonlinear feature interactions, while the learning ability of DL has improved significantly compared to shallow neural networks and machine learning models in general (Lara-Benítez et al. 2021).

Some research findings have shown the advantages of deep learning methods in prediction (Li et al. 2019b; Wu et al. 2021; Zhao et al. 2017), but they still have limitations for highly complex financial data. In order to further reason about the intricate temporal patterns and improve the forecasting accuracy, some researchers have considered time series decomposition methods (Abdollahi 2020; Li et al. 2021). Based on the idea of "divide and conquer," researchers decompose the time-series into several simple components and solve the original complex problems by modeling these components separately. The decomposition methods usually contain two categories. The first class of methods treats the time series as a superposition of several different intrinsic mode functions, mainly including wavelet transform-based decomposition, empirical mode decomposition (EMD) and variational mode decomposition (VMD). Another class of approaches that treat time series as a combination of trend, seasonal, and residual components includes STL (Cleveland et al. 1990) and TBATS (De Livera et al. 2011).

Decomposition-based deep learning models have been shown to have more accurate predictive performance in fields such as finance and environment (Dong et al. 2021; Sun et al. 2021). However, these time series decomposition methods usually employ one-time decomposition strategies (OTDS), that is, decomposition of all data containing both training data (historical data) and test data (future data) (Deng et al. 2020). In the context of forecasting, the efficiency of this preprocessing practice is limited in real-time forecasting scenarios because the future is unknown. In particular, the process of decomposing multiple series is tedious and time-consuming in a multiseries forecasting setting.

In this paper, we propose an end-to-end deep learning framework based on seasonal decomposition, which can be easily extended to various deep time series models. Our proposed framework supports probabilistic prediction functions, providing both parametric and non-parametric methods. Extensive empirical studies have shown that our framework compares favorably with state-of-the-art methods in both point prediction and probabilistic prediction tasks.

The four points considered in the proposed models compared to other models are as follows: (1) extremely fast series decomposition and fast training speed of the model; (2) high compatibility, as it is adaptable to various deep time series models; (3) nonlinear capability; and (4) compatibility with point prediction and probabilistic prediction.

The remainder of this study is as follows. Section 2 briefly reviews related literature on time series forecasting models, decomposition methods and their combination. Section 3 describes the framework we proposed. Section 4 investigates the forecasting performance of the models on different datasets. Section 5 gives the conclusions.

2 | Literature Review

Many research has applied deep learning methods in predictions. For example, Bristone et al. (2020) present a hybrid model for forecasting that uses complex network analysis and long short-term memory (LSTM). The experiments showed that the proposed model has higher accuracy and is more robust and reliable. Chiroma et al. (2015) proposed an alternative approach based on a genetic algorithm and neural network (GA-NN) for predictions. Based on the data transfer of prior knowledge, Cen and Wang (2019) empirically investigated the predictions effectiveness using LSTM, whose result showed that LSTM with data transfer can greatly improve the prediction accuracy of competing models. Yang et al. (2021b) achieved good results by using multiscale data and kernel extreme learning machine (KELM) to make predictions. Recently, based on complexity theory, Lin et al. (2022) made predictions combining LSTM with Google Trends (GT). Their results explained the weak ability and dynamic impact of GT in improving the long-term forecasting.

To reduce modelling difficulty, the researchers apply decomposition methods to transform the series into clear components and further model them with deep learning, which improve prediction performance. For example, Jammazi and Aloui (2012) combined Harr A trous wavelet decomposition and neural network model to make predictions, and they found that the hybrid model has higher efficiency and higher accuracy in crude oil forecasting. Li et al. (2018) proposed an EEMD-SBL-ADD model for forecastings by combining ensemble empirical mode decomposition (EEMD) and sparse Bayesian learning (SBL). Zhao et al. (2021) decomposed the series using an improved VMD method, whose parameters were optimized by the particle swarm according to the average maximum envelope entropy, and performed point prediction and interval prediction for the series. The interval prediction model was constructed based on the point prediction model and Bootstrap sampling. Recently, Guo et al. (2022) decomposed the multi-source series using VMD and fuzzy entropy (FE), then Granger causality test and random forest recursive feature elimination (RF-RFE) were used to censor useful predictor variables. Finally, each reconstruction mode was individually forecasted and the forecasting values obtained were combined and integrated. The empirical results showed that the proposed model improved the forecasting accuracy. More literature based on decomposition and ensemble forecasting frameworks can be found in (Salamai 2023; Sun et al. 2021; Sun et al. 2018).

Empirical studies have shown that the models can produce more accurate forecasts after de-seasonalization of the time series. For example, Makridakis et al. (1982) used 68 monthly time series of M-competition to compare the forecast performance produced by NNs trained with de-seasonalized and non-seasonalized data, and they found that the NNs trained with

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early de-seasonalization showed a significant improvement in forecasting accuracy. Yang et al. (2021a) proposed a Loess-based seasonal decomposition procedure combined with a one-dimensional convolutional neural network (STLeCNNePE) to predict the significant wave height, and their proposed model improved the prediction baseline. Zhang and Qi (2005)'s experiments showed that NNs are not effective in directly modeling trends or seasonality, and emphasized that prediction errors can be reduced by detrending or deseasonalizing the original time series. Taieb et al. (2012) and Bandara et al. (2020) confirmed these findings.

3 | Methodology

3.1 | Problem Definition

A w-horizon ahead probabilistic forecasting problem for time series can be described as follows. Given a set of time series $y_{1:T}$, our goal is to model the conditional distribution of the future observation, denoted as

$$P(y_{t+w}) = f(y_{t-k:t}, x_{t-k:t}), \tag{1}$$

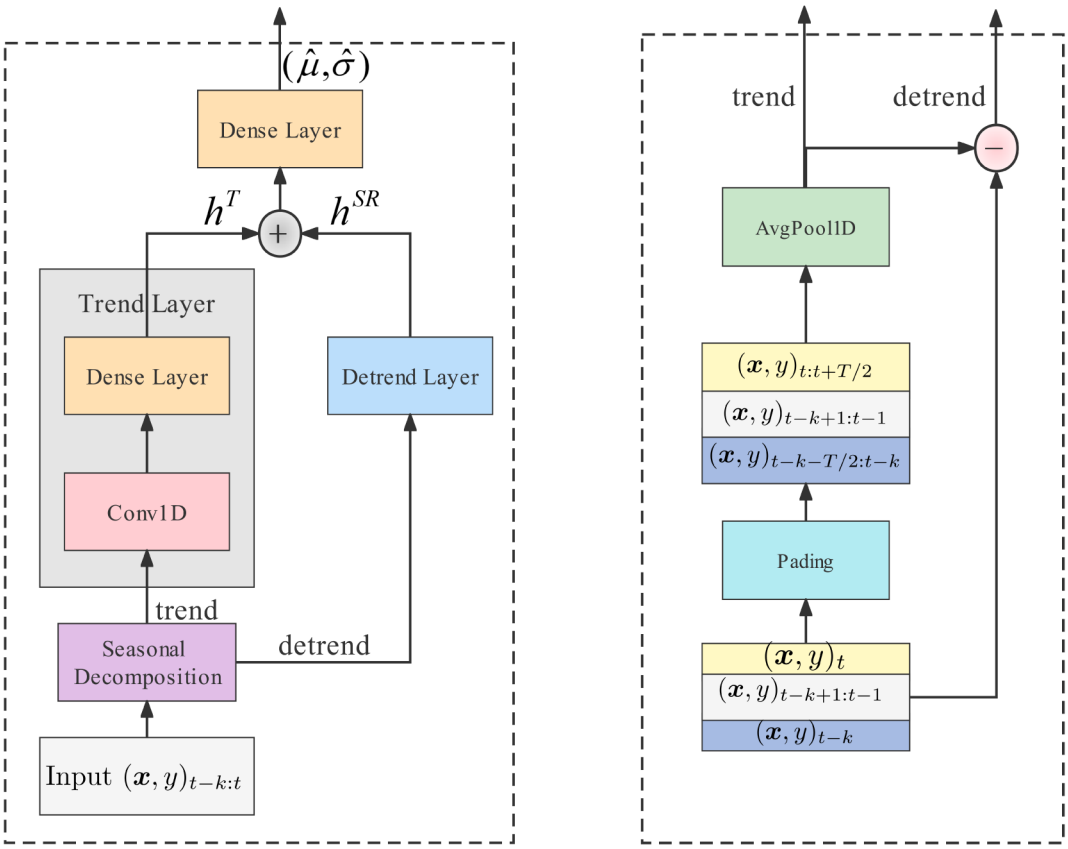
where $y_{t-k:t} = \{y_{t-k}, \ldots, y_t\}$ are observations of the target, $x_{t-k:t} = \{x_{t-k}, \ldots, x_t\}$ are exogenous inputs (covariates) that can only be observed historically over a look-back window and $f(\cdot)$ is the forecasting function learnt by the model. k is the length of the historical observations. In a broad sense, the deep neural networks used in time series forecasts are all sequence-to-sequence (seq2seq) structures, that is, the encoder is used to encode historical information into latent states h_t, and then the decoder is used to convert h_t and other information into outputs, namely:

$$\begin{aligned} h_t &= g_{\text{enc}}(y_{t-k:t}, x_{t-k:t}, s), \\ P(y_{t+w}) &= g_{\text{dec}}(h_t, y_{t-k:t}, x_{t-k:t}, s), \end{aligned} \tag{2}$$

where $g_{enc}(\cdot)$ and $g_{dec}(\cdot)$ are encoder and decoder, respectively. In a narrow sense, the model whose output is obtained only through the latent states h_t does not have a decoder module (generally transformed through a simple nonlinear layer), and the model at this time no longer has a seq2seq structure.

3.2 | Neural Network Architecture

In this paper, we propose an end-to-end deep learning framework based on seasonal decomposition for forecasting. The entire architecture of proposed framework is presented in Figure 1a. The high-level architecture contains four layers. include seasonal decomposition layer, detrend layer, trend layer and probabilistic forecasting layer. In the seasonal



(a) The proposed framework (b) Seasonal decomposition layer

FIGURE 1 | Architecture of the proposed framework.

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decomposition layer, the time series is first decomposed into a trend component and a detrend component, which is dominated by seasonal components. And this is why this framework is called seasonality-based decomposition. Subsequently, the trend layer and detrend layer are used to mine potential features for these two components respectively to obtain latent states h^T and h^{SR}. Finally, a dense layer is applied to map the states and to produce the probabilistic forecasts of future observations. In the following sections, we provide further details for each layer.

3.2.1 | Seasonal Decomposition Layer

Time series data can exhibit a variety of patterns. It is often helpful for prediction to decompose a time series into several components, each representing an underlying pattern category. In this paper, the time series is considered as a combination of trend component, seasonal component and residual component, and in the case of additive decomposition, we can write

$$y_t = S_t + T_t + R_t. \tag{3}$$

We assume that the residual component is a white noise, so the time series can be reduced to two components, trend component and detrend component, denoted as

$$y_t = T_t + SR_t. \tag{4}$$

In this paper, we apply a classical method of time series decomposition, a moving average method to estimate the trend component, which can be expressed as

$$\hat{T}_t = \frac{1}{m} \sum_{j=-k}^{k} y_{t+j}, \tag{5}$$

where $m = 2k + 1$. That is, the estimate of the trend at time t is obtained by averaging values of the time series within k periods. Observations that are nearby in time are also likely to be close in value. Therefore, the average eliminates some of the randomness in the data, leaving a smooth trend component. This is known as the $m - MA$ process, meaning m-order moving average (Dagum and Bianconcini 2016).

We apply a series decomposition layer to achieve real-time decomposition of the time series (see Figure 1b). The time series is first complemented with first and last values to avoid missing values at both ends of the series due to the $m - MA$ process, then the AvgPool layer is applied to simulate the $m - MA$ process by which we obtain the trend component, and finally the detrend component is obtained from the difference between the original series and the trend component.

3.2.2 | Detrend Layer

Four representative modules are used as detrend layers to mine the temporal and inter-variate interaction characteristics of the detrend components: two attention-based models, LogSparse Transformer (LogTrans) (Li et al. 2019a) and Temporal Pattern Attention (TPA) (Shih et al. 2019), one RNN-based model, Long Short-Term Memory network (LSTM) (Hochreiter and Schmidhuber 1997), and CNN-based Temporal Convolutional network (TCN) (Bai et al. 2018). With the detrend layer, the latent state of the detrend component h_t^{SR} can be obtained.

3.2.3 | Trend Layer

The Conv1D layer is first applied to smooth the trend components and mine the interaction features between variables. Then, the t-step outputs of the Conv1D layer are reduced to 1 using the dense layer, which is similar to the autoregressive model. With the trend layer, latent state of the trend component h_t^T can be obtained.

3.2.4 | Probabilistic Forecasting Layer

We consider two probabilistic forecasting methods in this paper. The first one is a parametric method. It requires a preassumed probability distribution (e.g., Gaussian distribution), based on the negative log-likelihood loss, to achieve probabilistic forecasts of future observations. Since it directly forecasts the parameters of the assumed distribution, it allows flexibility to obtain additional information by sampling from the estimated distribution. The second one is a nonparametric method. Based on the quantile loss, it allows the model to output a set of forecasts corresponding to the quantile points of interest. This method is distribution-free and therefore usually more robust. Specifically, in the probabilistic forecasting layer, h_t^T and h_t^{SR} are first summed as the latent output of the historical time series, and then, the dense layer is used to generate m outputs $Z = (h_1, \cdots, h_m)$, which represent the parameters of the hypothetical distribution of interest.

For the parametric method, we assume that the time series follows a Gaussian distribution, and for the future observation y_{t+1}, the probabilistic forecasting layer produces $H_{t+1} = (\mu_{t+1}, \sigma_{t+1})$, where μ_{t+1} is the expected value and σ_{t+1} is the standard deviation of y_{t+1}. Then, the negative log-likelihood function is constructed as a loss function:

$$\begin{aligned} L_G &= -\log \ell(\mu, \sigma | y) \\ &= -\log\left(\left(2\pi\sigma^2\right)^{-1/2} \exp\left[-(y-\mu)^2/\left(2\sigma^2\right)\right] \right) \\ &= \frac{1}{2}\log(2\pi) + \log(\sigma) + \frac{(y-\mu)^2}{2\sigma^2}. \end{aligned} \tag{6}$$

In addition, to ensure that the standard deviation of the probabilistic forecasting layer satisfies the positivity condition (Chen et al. 2020; Salinas et al. 2020), we use the SoftReLU activation function for the standard deviation, which is denoted as $z = log(1 + exp(z))$.

For the nonparametric method, forecasts can be obtained by quantile regression. The observation and forecasts of a specific quantile level q are denoted as y and $\hat{y}$, respectively, and the q-quantile is defined as

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$$L_q(y, \hat{y}, q) = q(y - \hat{y})^+ + (1 - q)(\hat{y} - y)^+, \tag{7}$$

where $(\cdot)^+ = max(0, \cdot)$ and $q \in (0, 1)$. Model is trained by jointly minimizing the quantile loss, summed across all quantile outputs:

$$L_Q = \sum_{y_t \in \Omega} \sum_{q \in \mathcal{Q}} \frac{L_q(y_t, \hat{y}, q)}{M}, \tag{8}$$

where Ω is the domain of training data containing M samples and $\mathcal{Q} = \{0.1, 0.5, 0.9\}$ is the set of output quantiles (Lim et al. 2021).

3.3 | Performance Evaluation Criteria

For out-of-sample testing, we evaluate the normalized quantile losses across the entire forecasting horizon-focusing on P50 risk, which is denoted as

$$q\text{-}Risk = \frac{\sum_{y_t \in \Omega} L_q(y_t, \hat{y}_t, q)}{\sum_{y_t \in \Omega} |y_t|}, \tag{9}$$

where $L_q(\cdot)$ is defined in Equation (7), Ω is the domain of test data, and $q = 0.5$. Root mean square error (RMSE) is used in our experiments for point forecasting, which is denoted as

$$RMSE = \sqrt{\frac{1}{N} \sum_t (y_t - \hat{y}_t)^2}, \tag{10}$$

where y_t is the true value at time step t and $\hat{y}_t$ is the corresponding prediction value.

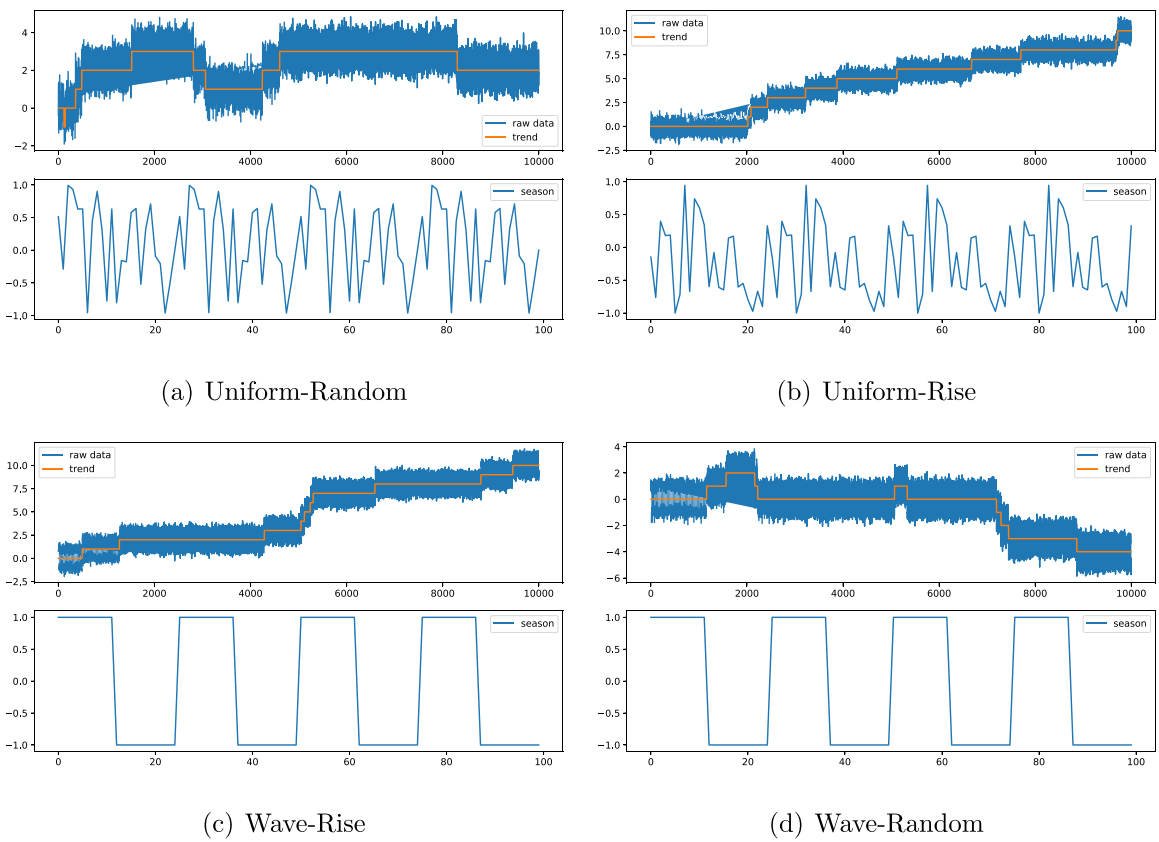
3.4 | Datasets

3.4.1 | Synthetic Datasets

To generate the synthetic dataset, we incorporate different types of seasonality, trend changes, and noise to simulate real-world scenarios, as shown in Figure 2. Specifically, we generate two types of seasonal components. For a periodic signal, the first one (Uniform) follows a uniform distribution $y_t \sim U(-1, 1)$, and the second one (Wave) has a square wave-like shape, which is denoted as

$$y = \begin{cases} -1, & t < \frac{T}{2} \\ +1, & t \geqslant \frac{T}{2} \end{cases} \tag{11}$$

The seasonal period T is set to 25, and a total of 10,000 time points are generated. In addition, this paper randomly selects 10 moments and randomly rises or falls by 1 unit at the corresponding moment to form the first trend component (Random) and rises by 1 unit at each selected moment to form the second trend component (Rise). Four synthetic datasets are generated by combining seasonal and trend components two-by-two (see Figure 2). The noise is added by the standard normal distribution.



(a) Uniform-Random (b) Uniform-Rise

(c) Wave-Rise (d) Wave-Random

FIGURE 2 | Generated synthetic data where the top subplot represents the raw data and the trend, and the bottom subplot represents the seasonality.

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3.4.2 | WTI Crude Oil Price Dataset

For this study, the daily WTI crude oil spot price series for the period from 1986 to 2023 were extracted from (EIA 2024) to construct the single-series crude oil price dataset (WTI-S). In addition, considering that stock market, exchange rate market, and global economic activities may affect the crude oil prices (Wang et al. 2015; Zhao et al. 2017), we collect the CBOE DJIA Volatility Index (VIX), the US Economic Policy Index of Uncertainty (EPU), and the

TABLE 1 | Descriptive statistics for the WTI crude oil price dataset.

	WTI	VIX	EPU	TWUSDX
Mean	57.45	19.02	112.16	112.75
Std	28.26	7.77	35.42	10.10
Min	10.82	0.00	57.20	93.96
Max	145.31	74.60	245.13	131.58
Skew	0.35	1.81	0.84	−0.04
Kurt	−0.75	5.71	0.21	−1.22
JB	242.73***	10,531.70***	664.86***	343.33***
Q(5)	27,530.18***	24,798.05***	26,093.12***	27,616.63***
Q(10)	54,756.34***	47,317.14***	49,574.01***	55,026.88***
Q(20)	108,273.77***	87,475.52***	89,239.86***	109,116.25***
ADF	−1.94	−4.66***	−4.48***	−0.98

TABLE 2 | P50 performance of parametric probabilistic forecasting on synthetic datasets.

Dataset	Horizon	LSTM	LSTM-De	TPA	TPA-De	TCN	TCN-De	LogTrans	LogTrans-De	Count
Uniform-Random	1	0.134	0.066	**0.062**	**0.061**	0.066	0.066	0.081	0.080	4
	3	0.061	**0.059**	0.068	**0.060**	0.067	0.065	0.070	0.063	4
	6	**0.062**	**0.061**	0.069	0.062	0.066	0.063	0.075	0.063	4
	10	0.138	**0.061**	0.068	**0.061**	0.074	0.065	0.087	0.068	4
	Avg	0.091	**0.061**	0.066	**0.062**	0.070	0.065	0.076	0.088	3
Uniform-Rise	1	0.061	0.022	0.054	**0.019**	0.022	**0.019**	0.051	0.262	3
	3	0.065	**0.017**	0.037	**0.018**	0.025	0.019	0.052	0.036	4
	6	0.048	0.020	0.049	**0.018**	0.022	**0.019**	0.049	0.035	4
	10	0.057	**0.018**	0.070	**0.018**	0.028	0.021	0.044	0.031	4
	Avg	0.057	**0.018**	0.055	**0.018**	0.023	0.019	0.048	0.055	3
Wave-Rise	1	0.030	**0.016**	0.025	0.019	0.020	**0.016**	0.040	0.020	4
	3	0.046	0.017	0.045	0.017	0.027	**0.016**	0.041	**0.017**	4
	6	0.054	0.017	0.073	**0.016**	0.032	**0.017**	0.039	0.025	4
	10	0.042	0.020	0.083	**0.017**	0.027	**0.017**	0.034	0.018	4
	Avg	0.049	0.018	0.047	**0.018**	0.029	**0.017**	0.043	0.018	4
Wave-Random	1	0.569	**0.152**	0.469	0.297	0.603	**0.109**	0.392	0.245	4
	3	0.635	0.337	0.629	**0.250**	0.581	0.276	0.363	**0.256**	4
	6	0.622	**0.333**	0.643	**0.294**	0.611	0.357	0.359	0.370	3
	10	0.643	0.319	0.612	0.319	0.706	**0.267**	0.515	**0.311**	4
	Avg	0.622	**0.282**	0.572	0.315	0.627	**0.245**	0.438	0.348	4

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trade-weighted dollar index (TWUSDX) to construct the multiseries crude oil price dataset (WTI-MS) for illustrating the compatibility of our proposed framework for multiseries modeling (Guo et al. 2022).

Table 1 shows the descriptive statistics for the WTI crude oil price dataset, from which it can be seen that WTI crude oil prices have a positive skewness (0.35) and a kurtosis value of –0.75, and the Jarque-Bera statistic rejects the hypothesis of normal distribution at 1% confidence level. The Ljung-Box test results of the 5th-, 10th-, and 20th-order lagged terms indicate that oil prices have significant autocorrelation characteristics, which indicates the long memory nature in the crude oil spot market. The ADF statistic indicates that series is not smooth.

4 | Results

We selected the four models mentioned in Section 3.2.2 as the baseline and used them to forecast the target values for the next 10 moments. In addition, to distinguish from these baseline methods, we wrote the series decomposition-based model as Model-De. All datasets were divided according to 0.7:0.1:0.2 without disrupting the chronological order. All models were trained using the Adam optimizer (Kingma and Ba 2014) with an initial learning rate of 10*e*-3. Batch size was set to 32. The training process was early stopped within 50 epochs. All experiments were repeated three times, implemented in PyTorch.

4.1 | Results on Synthetic Datasets

For the synthetic dataset, we performed forecasting for 10 future time points using the series length of the last 50 (containing two periods) and evaluated the probabilistic forecasting performance of our proposed framework using the q-risk of 0.5 and 0.9 quantiles. It should be noted that for the parametric probabilistic forecasting model, Gaussian distributions corresponding to the forecasting parameters (mean and standard deviation) were generated and used to compute the 0.5 and 0.9 quantile forecasting values. In addition, we evaluated the point forecasting performance using RMSE.

Tables 2, 3, and 4 show the parametric probability, nonparametric probability, and point prediction performance of the model on the four synthetic datasets, respectively. We label the top

TABLE 3 | P50 performance of nonparametric probabilistic forecasting on synthetic datasets.

Dataset	Horizon	LSTM	LSTM-De	TPA	TPA-De	TCN	TCN-De	LogTrans	LogTrans-De	Count
Uniform-Random	1	**0.059**	0.077	0.069	**0.063**	0.068	0.063	0.086	0.076	3
	3	0.065	**0.061**	**0.064**	0.066	0.071	0.065	0.069	0.130	2
	6	0.116	**0.060**	0.068	**0.065**	0.070	0.065	0.083	0.113	3
	10	0.132	**0.061**	0.065	**0.064**	0.069	0.066	0.067	0.074	3
	Avg	0.095	0.069	0.067	**0.064**	0.071	**0.065**	0.076	0.105	3
Uniform-Rise	1	0.052	**0.020**	0.025	**0.017**	0.023	0.020	0.049	0.024	4
	3	0.052	**0.017**	0.030	**0.018**	0.023	0.018	0.055	0.036	4
	6	0.043	0.020	0.033	**0.017**	0.027	**0.018**	0.048	0.037	4
	10	0.066	**0.018**	0.069	**0.018**	0.051	0.019	0.038	0.037	4
	Avg	0.057	0.022	0.045	**0.019**	0.028	**0.020**	0.047	0.033	4
Wave-Rise	1	0.035	0.017	0.023	**0.015**	0.023	**0.016**	0.039	0.018	4
	3	0.043	0.019	0.059	**0.017**	0.031	**0.016**	0.031	0.017	4
	6	0.041	0.018	0.033	**0.018**	0.036	**0.017**	0.039	0.020	4
	10	0.065	0.017	0.048	0.017	0.025	**0.017**	0.046	**0.016**	4
	Avg	0.048	0.018	0.041	0.018	0.032	**0.017**	0.038	**0.017**	4
Wave-Random	1	0.623	0.125	0.315	**0.117**	0.538	0.296	0.364	**0.113**	4
	3	0.608	0.222	0.397	0.331	0.566	**0.149**	0.419	**0.161**	4
	6	0.635	**0.295**	0.571	0.408	0.585	**0.258**	0.760	0.324	4
	10	0.658	0.630	0.763	0.297	0.688	**0.167**	0.543	**0.119**	4
	Avg	0.620	**0.244**	0.562	0.277	0.611	**0.259**	0.518	0.276	4

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TABLE 4 | RMSE performance of point forecasting on synthetic datasets.

Dataset	Horizon	LSTM	LSTM-De	TPA	TPA-De	TCN	TCN-De	LogTrans	LogTrans-De	Count
Uniform-Random	1	0.405	0.396	0.345	**0.344**	0.381	**0.345**	0.370	0.407	3
	3	0.346	**0.334**	0.363	**0.341**	0.376	0.361	0.401	0.400	4
	6	0.416	**0.340**	0.367	**0.346**	0.384	0.354	0.371	0.357	4
	10	0.462	**0.339**	0.360	**0.349**	0.386	0.353	0.394	0.373	4
	Avg	0.382	0.356	0.358	**0.346**	0.381	**0.354**	0.382	0.406	3
Uniform-Rise	1	1.029	**0.359**	0.542	**0.347**	0.418	0.362	0.803	0.396	4
	3	1.326	**0.344**	0.501	**0.353**	0.431	0.372	0.927	0.688	4
	6	0.810	**0.348**	1.101	**0.356**	0.464	0.359	0.847	0.371	4
	10	1.227	0.395	2.103	**0.357**	0.469	**0.381**	0.862	0.402	4
	Avg	1.035	**0.358**	0.987	**0.359**	0.455	0.371	0.857	0.469	4
Wave-Rise	1	0.898	0.365	0.516	**0.348**	0.452	**0.349**	0.928	0.352	4
	3	0.998	**0.350**	0.489	**0.349**	0.593	0.352	0.750	0.357	4
	6	0.761	0.361	0.666	0.357	0.594	**0.349**	0.904	**0.348**	4
	10	0.880	0.386	1.151	**0.353**	0.592	**0.363**	0.628	0.363	4
	Avg	0.953	0.372	0.853	**0.356**	0.657	**0.352**	0.842	0.360	4
Wave-Random	1	3.515	0.344	2.147	**0.338**	2.804	**0.337**	2.680	0.339	4
	3	3.974	0.366	3.257	**0.348**	3.162	**0.338**	2.673	0.423	4
	6	4.604	**0.340**	4.611	0.361	3.481	**0.346**	2.611	0.379	4
	10	4.400	0.392	1.164	**0.343**	3.791	**0.350**	2.676	0.385	4
	Avg	4.154	0.366	2.939	**0.347**	3.519	**0.348**	2.722	0.370	4

TABLE 5 | P50 performance of parametric probabilistic forecasting on WTI dataset.

Dataset	Horizon	LSTM	LSTM-De	TCN	TCN-De	TPA	TPA-De	LogTrans	LogTrans-De	Count
WTI-MS	1	0.030	**0.009**	0.034	0.010	0.017	**0.010**	0.048	0.014	4
	3	0.033	**0.014**	0.032	0.015	0.022	**0.015**	0.031	0.021	4
	6	0.037	**0.023**	0.045	**0.021**	0.046	0.023	0.028	0.027	4
	10	0.048	**0.027**	0.078	0.029	0.068	**0.027**	0.068	0.042	4
	Avg	0.039	**0.019**	0.048	**0.019**	0.049	0.020	0.050	0.030	4
WTI-S	1	0.011	**0.009**	0.010	0.009	0.017	**0.008**	0.016	0.014	4
	3	0.017	0.015	0.015	**0.014**	0.015	**0.015**	0.024	0.019	4
	6	0.022	**0.020**	0.022	0.022	0.022	**0.020**	0.022	0.024	3
	10	0.028	**0.027**	0.029	**0.027**	0.027	0.028	0.033	0.027	3
	Avg	0.022	**0.019**	0.020	0.019	0.022	**0.019**	0.027	0.022	4

two models with red and green that are predicted to perform well. The *Count* column indicates the number of times our proposed framework is better compared to the original model, with a maximum of 4. Observing Tables 2 and 3, we can find that the seasonal decomposition has a significant improvement on probabilistic prediction ability of deep learning, as shown by the fact that the *Count* column is mostly 4. In the case of LSTM, for example, this high complexity makes it perform worse

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on Wave-Random data than on other synthetic datasets due to the obvious difference between the train and test sets (see Figure 2d). Even so, the parametric and nonparametric probabilistic predictive power of LSTM-De based on seasonal decomposition can be maintained above 50% on average.

Table 4 shows the performance of the point forecasts. It can be found that, similar to the results for probabilistic forecasting, the seasonal decomposition-based models perform better at almost all forecast moments for all synthetic datasets.

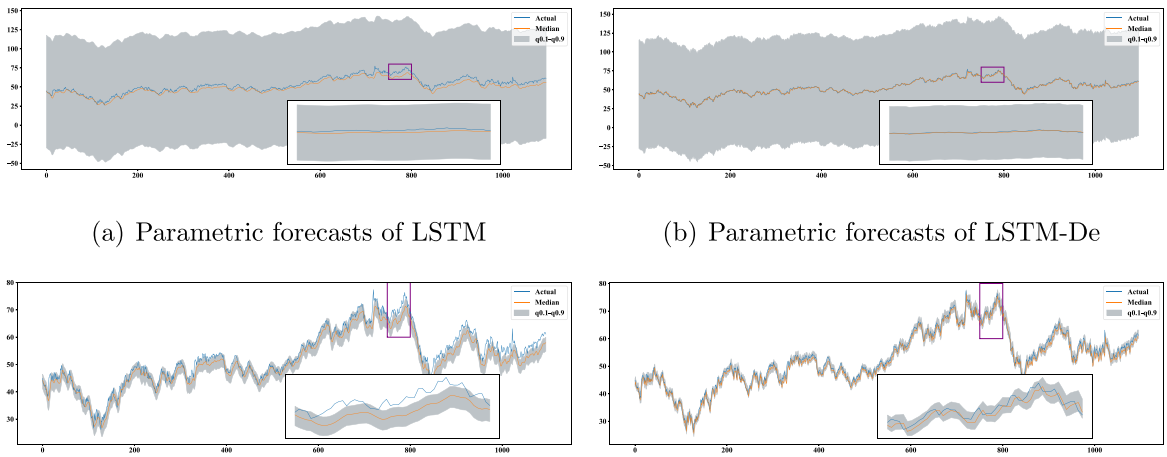
4.2 | Results on WTI Crude Oil Price Dataset

The probabilistic forecasting results of WTI-MS and WTI-S are shown in Tables 5 and 6. We can find that the seasonal decomposition-based models achieve better forecasting results compared to the original models. From the forecasting performance of the 0.5 quantile, the nonparametric probabilistic models do not show significant dominance over the parametric probabilistic model. However, the forecasting results for the 0.9 quantile show the superiority of the nonparametric approach. For presentation purposes, the results for the 0.9 quantile are not given, and a supporting evidence can be seen from Figure 3. One possible reason for this is that parametric probabilistic forecasting models are constructed based on Gaussian distributions, but as Table 1 shows, the WTI crude oil price does not follow this assumption. In contrast, the nonparametric probabilistic forecasting models, which take into account the distribution-free nature, produce better forecasts by directly minimizing quantile loss.

To better understand the performance improvement of the proposed framework, we show the probabilistic forecasting results of the LSTM and LSTM-De for the WTI-MS dataset a-day ahead in Figure 3, where the top and the bottom subplot are the parametric and non-probabilistic results, respectively. Comparing the nonparametric probabilistic performance of LSTM and LSTM-De, see Figure 3c,d, we can find that the target value is always within the 0.1 and 0.9 quantile prediction range of LSTM-De, while the prediction interval of LSTM can only track the trend of the target value. In addition, comparing the results

TABLE 6 | P50 performance of nonparametric probabilistic forecasting on WTI dataset.

Dataset	Horizon	LSTM	LSTM-De	TCN	TCN-De	TPA	TPA-De	LogTrans	LogTrans-De	Count
WTI-MS	1	0.020	**0.011**	0.015	**0.009**	0.022	0.012	0.015	0.013	4
	3	0.047	0.017	0.032	**0.015**	0.042	**0.014**	0.029	0.021	4
	6	0.041	**0.020**	0.023	**0.020**	0.070	0.025	0.043	0.024	4
	10	0.043	**0.027**	0.046	0.032	0.068	**0.029**	0.058	0.030	4
	Avg	0.041	**0.019**	0.038	**0.020**	0.058	0.021	0.041	0.023	4
WTI-S	1	0.013	**0.009**	0.010	0.009	**0.009**	0.009	0.039	0.015	3
	3	0.018	**0.014**	0.015	**0.015**	0.016	0.015	0.027	0.016	4
	6	0.022	0.020	0.021	**0.020**	0.022	**0.020**	0.028	0.022	4
	10	0.028	**0.026**	0.027	**0.026**	0.029	0.027	0.032	0.030	4
	Avg	0.021	0.019	0.020	**0.019**	0.021	**0.019**	0.029	0.023	4



(a) Parametric forecasts of LSTM

(b) Parametric forecasts of LSTM-De

(c) Nonparametric forecasts of LSTM

(d) Nonparametric forecasts of LSTM-De

FIGURE 3 | Parametric and nonparametric probabilistic forecasts of LSTM-type for 1 horizon ahead on WIT-MS dataset.

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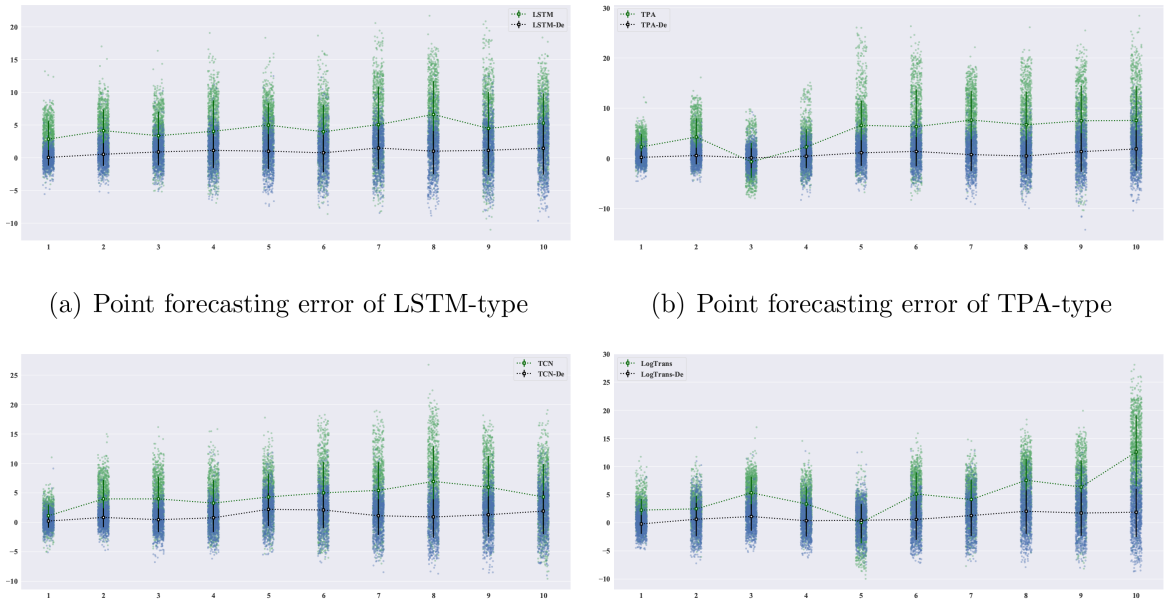
TABLE 7 | RMSE performance of point forecasting on WTI dataset.

Dataset	Horizon	LSTM	LSTM-De	TCN	TCN-De	TPA	TPA-De	LogTrans	LogTrans-De	Count
WTI-MS	1	3.801	**1.235**	2.147	**1.237**	3.224	1.269	3.166	1.711	4
	3	4.702	2.180	5.225	**2.027**	3.486	**1.935**	6.050	2.548	4
	6	5.730	**2.883**	7.140	3.652	9.087	**3.162**	6.274	3.196	4
	10	6.814	**3.802**	6.613	**4.023**	9.679	4.349	13.800	4.408	4
	Avg	6.166	**2.890**	6.277	3.002	7.627	**2.969**	6.956	3.284	4
WTI-S	1	2.123	**1.210**	1.360	**1.223**	1.315	1.240	2.309	1.687	4
	3	2.377	**2.009**	2.196	2.246	2.246	**1.918**	2.545	2.526	3
	6	2.850	3.035	**2.823**	2.827	2.869	**2.698**	3.267	3.057	2
	10	3.700	3.612	3.599	**3.508**	3.762	**3.509**	4.377	3.934	4
	Avg	2.937	2.672	2.735	**2.667**	3.229	**2.621**	3.182	3.081	4



(a) Point forecasting error of LSTM-type

(b) Point forecasting error of TPA-type

(c) Point forecasting error of TCN-type

(d) Point forecasting error of LogTrans-type

FIGURE 4 | Point forecasting error distribution for 10 horizons ahead on WIT-MS dataset.

of probability prediction based on Gaussian distribution with those based on quantile loss, the results of nonparametric probability prediction are more accurate, and this finding is consistent with the performance of the models on synthetic datasets.

Table 7 shows the point forecasting performance of the model on the WTI-MS and WTI-S datasets. As can be seen, our proposed seasonal decomposition-based framework is still valid, and it improves the forecasting performance of almost all baseline models. In addition, there is no accuracy gain in WTI crude oil price forecasting by adding additional exogenous variables. To illustrate the high performance of the proposed framework, we plot the point forecasting error distribution of the models for the WTI-MS dataset in Figure 4. It can be found that the performance of the models become worse as the predicted moment is extended, and the error distribution of Model-De is around 0, while the corresponding original model error deviates from 0 and its dispersion is higher.

5 | Conclusion

We present an end-to-end deep learning framework based on seasonal decomposition for probabilistic forecasting that supports both nonparametric and parametric methods to model probabilistic distributions. Distinguishing from models based on time series pre-decomposition, our proposed framework is end-to-end. And since it models the trend component and the seasonal component separately, the framework can be adapted to various deep time series models.

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To test the forecasting ability of the proposed framework, we first construct four synthetic datasets containing different seasonal and trend components to verify the compatibility of the framework for forecasting different types of time series. On this basis, we further use it to forecast the WTI crude oil prices.

The results of five datasets show that the deep time series model based on seasonal decomposition yields superior performance compared to original neural network model in both point and probabilistic forecasting.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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